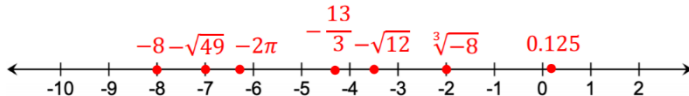


## 2.5 Rational and Irrational Numbers on the Number Line

### Lesson Introduction

|                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                               |                                                                                                                                          |                                                                                                                                                                                                                        |
|-------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  Benchmark           | MA.8.NSO.1.2 Plot, order, and compare rational and irrational numbers, represented in various forms.                                                                                                                                                                                                                                                                                                                               |                                                                                               |  Learning Targets                                     | <ul style="list-style-type: none"><li>I can compare rational and irrational values.</li><li>I can order rational and irrational values.</li><li>I can plot rational and irrational numbers on a number line.</li></ul> |
|  Benchmark Coherence | Building On                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                               |                                                                                                                                          | Working Toward                                                                                                                                                                                                         |
|                                                                                                       | MA.7.NSO.1.2 Rewrite rational numbers in different but equivalent forms, including fractions, mixed numbers, repeating decimals, and percentages, to solve mathematical and real-world problems.                                                                                                                                                                                                                                   |                                                                                               |                                                                                                                                          | 912.NSO1.4 Apply previous understanding of operations with rational numbers to add, subtract, multiply, and divide numerical radicals.                                                                                 |
|  Essential Questions | <ul style="list-style-type: none"><li>How do you compare rational and irrational values?</li><li>How do you order rational and irrational values?</li><li>How do you plot rational and irrational numbers on a number line?</li></ul>                                                                                                                                                                                              |                                                                                               |                                                                                                                                          |                                                                                                                                                                                                                        |
|  Lesson Narrative    | This lesson builds on students' knowledge of and skills with approximating and determining square and cube roots as well as expressions including pi. Students compare rational and irrational numbers using $<$ and $>$ symbols and order them by representing them on a number line. The lesson includes an activity where students work together to order and plot assigned numbers on designated number lines around the room. |                                                                                               |                                                                                                                                          |                                                                                                                                                                                                                        |
|  Component Summary | 2.5.1 Warm-Up (~5 min.)                                                                                                                                                                                                                                                                                                                                                                                                            | Determine which number doesn't belong ( <i>SE p. 87</i> )                                     | 2.5.4 Activity (10-15 min.)                                                                                                              | Identify numbers as rational or irrational and work as a group to order and plot on a number line ( <i>SE p. 89</i> )                                                                                                  |
|                                                                                                       | 2.5.2 Guided Instruction (10-15 min.)                                                                                                                                                                                                                                                                                                                                                                                              | Compare, order, and plot rational and irrational numbers on a number line ( <i>SE p. 88</i> ) | 2.5.5 Bring It Together (5 min.)                                                                                                         | Discuss strategies for comparing, ordering, and plotting numbers on a number line ( <i>SE p. 90</i> )                                                                                                                  |
|                                                                                                       | 2.5.3 Guided Practice (5 min.)                                                                                                                                                                                                                                                                                                                                                                                                     | Order numbers from least to greatest and plot on a number line ( <i>SE p. 89</i> )            | 2.5.6 Wrap-Up (~5 min.)                                                                                                                  | Students summarize their learning ( <i>SE p. 90</i> )                                                                                                                                                                  |
|  Resources         | Glossary                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                               | Materials                                                                                                                                |                                                                                                                                                                                                                        |
|                                                                                                       | <u>Key Terms:</u> N/A<br><u>Additional Terms:</u> N/A                                                                                                                                                                                                                                                                                                                                                                              |                                                                                               | <ul style="list-style-type: none"><li>Print and cut out activity cards for 2.5.4 Activity</li><li>Copies of blank number lines</li></ul> |                                                                                                                                                                                                                        |
|                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                               |                                                                                                                                          | Additional Preparation                                                                                                                                                                                                 |
|                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                               |                                                                                                                                          | <ul style="list-style-type: none"><li>(Optional) Prepare two signs: "Rational" and "Irrational" for 2.5.4 Activity.</li></ul>                                                                                          |

|                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                         |
|---------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <br>Teacher Tips | <b>Common Misconceptions</b> <ul style="list-style-type: none"><li>Students may divide by 2 instead of finding the square root or divide by 3 instead of finding the cube root.</li><li>Students may know which value is greater but may be confused about which inequality symbol to use.</li></ul>                                                                                                                                                                                                              | <b>Things to Consider</b> <ul style="list-style-type: none"><li>Consider creating a classroom number line and having students plot points and justify their placement of the points to debrief the lesson.</li></ul>                                                                                                                    |
|                                                                                                   | <b>Literacy and Language Routines</b> <p>Which One Doesn't Belong?<br/>In 2.5.1 Warm-Up, students explain why one of the four numbers does not belong with the other three. Consider prompting students to revise and refine their explanations to ensure they are precise and accurate.</p>                                                                                                                                                                                                                      | <b>Mathematical Thinking and Reasoning (MTR) Standard</b> <p><b>MA.K12.MTR.4.1: Discussion and Reflection</b><br/>In 2.5.4 Activity, students work together to identify numbers as rational or irrational and then order and plot the numbers on a number line. Students justify their reasoning and analyze each other's thinking.</p> |
|                                                                                                   | <p>In 2.5.4 Activity, students are provided with a kinesthetic entry point when they sort rational and irrational numbers and move to the designated side of the room. Creating a classroom number line and asking students to justify their placement of values provides visual, kinesthetic, and audio entry points.</p> <div></div> <p>Additional differentiation opportunities are denoted throughout the lesson guide.</p> |                                                                                                                                                                                                                                                                                                                                         |

| Rational Numbers | Irrational Numbers |
|------------------|--------------------|
| 0.125            | $-\sqrt{12}$       |
| -8               | $-2\pi$            |
| $-\sqrt{49}$     |                    |
| $\sqrt[3]{-8}$   |                    |
| $-\frac{13}{3}$  |                    |

Lesson Notes:

## 2.5.1 Warm-Up: Which One Doesn't Belong?

**Component Overview:** Students determine which number doesn't belong with the other three. Students will justify why their selection doesn't belong with the others. Students are asked to make an argument for why two of the values don't belong.



### 2.5.1 Warm-Up: Which One Doesn't Belong?

- Four numbers are shown in the table. Choose **two** of the values to write an argument for why each does not belong with the others.

*Answers will vary.*

| Number            | Why doesn't this belong with the others?                                                                                                                   |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3.14              | <i>This number does not belong with the others because it is the only terminating decimal.</i>                                                             |
| $\frac{22}{7}$    | <i>This number does not belong with the others because it expressed as a fraction and the others are not.</i>                                              |
| $\pi$             | <i>This number does not belong with the others because it is the only irrational number.</i>                                                               |
| $3.\overline{14}$ | <i>This number does not belong with the others because the decimal notation repeats. It is also the only number that is not pi or used to approximate.</i> |



#### Which One Doesn't Belong?

Students explain why one of the four numbers doesn't belong. Consider prompting students to revise and refine their explanations to ensure they are precise and accurate. An argument could be made for any of the values.

## 2.5.2 Guided Instruction: Compare and Order Rational and Irrational Numbers

**Component Overview:** Students sort numbers as either rational or irrational. Then they compare and order numbers and plot them on a number line.



### 2.5.2 Guided Instruction: Compare and Order Rational and Irrational Numbers

1. Use the numbers shown to complete the following.

$$0.125 \quad -\sqrt{12} \quad -8 \quad -\sqrt{49} \quad -2\pi \quad \sqrt[3]{-8} \quad -\frac{13}{3}$$

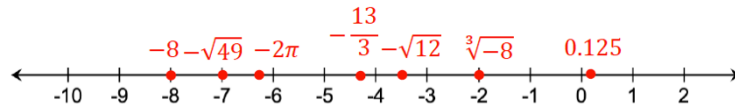
- a. Write each number under the appropriate description in the table.

| Rational Numbers | Irrational Numbers |
|------------------|--------------------|
| 0.125            | $-\sqrt{12}$       |
| -8               | $-2\pi$            |
| $-\sqrt{49}$     |                    |
| $\sqrt[3]{-8}$   |                    |
| $-\frac{13}{3}$  |                    |

- b. Order the numbers from least to greatest.

$$-8; -\sqrt{49}; -2\pi; -\frac{13}{3}; -\sqrt{12}; \sqrt[3]{-8}; 0.125$$

- c. Plot and label each number on the number line.



Model the thinking and reasoning used to order the values from least to greatest and plot them on the number line.

## 2.5.2 Guided Instruction: Compare and Order Rational and Irrational Numbers (continued)

2. Complete the table by comparing each pair of expressions using  $>$ ,  $<$ , or  $=$ . Then, plot and label the expressions on the number line.

| Compare<br>( $>$ , $<$ , or $=$ )                                                                                                                                      | Number Line Plot |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| $\sqrt{36}$ <div> <input type="radio"/> <math>&lt;</math><br/> <input type="radio"/> <math>&gt;</math><br/> <input type="radio"/> <math>=</math> </div> $\sqrt[3]{36}$ |                  |
| $\sqrt{89}$ <div> <input type="radio"/> <math>&lt;</math><br/> <input type="radio"/> <math>&gt;</math><br/> <input type="radio"/> <math>=</math> </div> $9.89$         |                  |
| $\frac{9}{2}$ <div> <input type="radio"/> <math>&lt;</math><br/> <input type="radio"/> <math>&gt;</math><br/> <input type="radio"/> <math>=</math> </div> $\sqrt{23}$  |                  |
| $\pi + 3$ <div> <input type="radio"/> <math>&lt;</math><br/> <input type="radio"/> <math>&gt;</math><br/> <input type="radio"/> <math>=</math> </div> $2\pi$           |                  |



Based on formative data and observations during the previous lesson, determine whether it's appropriate to release students to try question 2 with a partner or whether additional modeling is needed. Consider also releasing partners and then still pulling a small group of students who want or need additional support before trying on their own.

If question 2 is released to students, review the answers prior to moving to the next component to address misconceptions, answer questions, and ensure stronger understanding.

## 2.5.3 Guided Practice: Compare and Order Rational and Irrational Numbers

**Component Overview:** Students practice ordering a set of rational and irrational numbers and plotting them on a number line.



### 2.5.3 Guided Practice: Compare and Order Rational and Irrational Numbers

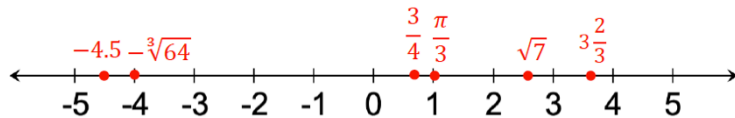
1. Use the numbers shown to complete the following.

$$-4.5 \quad \frac{3}{4} \quad \sqrt{7} \quad 3\frac{2}{3} \quad -\sqrt[3]{64} \quad \frac{\pi}{3}$$

- a. Order the numbers from greatest to least.

$$3\frac{2}{3}; \sqrt{7}; \frac{\pi}{3}; \frac{3}{4}; -\sqrt[3]{64}; -4.5$$

- b. Plot and label each number on the number line.



Allow students to work with a partner to help them build their confidence with the skills. Prompt students to discuss how they approximated the nonperfect roots and how they determined where to plot the values on the number line.



Monitor students as they're working to determine if there are any values that need to be discussed whole group.

## 2.5.4 Activity: Ordering Rational and Irrational Numbers

**Component Overview:** Students are given a card with a numerical expression on it. First, they determine if the expression is rational or irrational and move to the designated side of the classroom. Next, they collaborate to compare and order their values on a number line. Refer to the 2.5.4 Activity document for details on implementing this activity.



### 2.5.4 Activity: Ordering Rational and Irrational Numbers

1. Your teacher will give you a card with a numerical expression on it.
  - a. What is the numerical expression on your card?  
*Answers will vary. Refer to activity document.*
  - b. Does the expression include any irrational numbers?  
*Answers will vary. Refer to activity document.*
  - c. If your expression contains only rational numbers, write the exact value of your card, expressed as a decimal. If your expression contains any irrational numbers, write the approximate value of your card, expressed as a decimal.  
*Answers will vary. Refer to activity document.*
2. Once in your group, label the tick marks on the number line to match your group's assigned number line.



- a. Work with your group to order your number cards from least to greatest.  
*Answers will vary. Refer to activity document.*
- b. Work with your group to determine the approximate location of each expression on the number line. Then, plot and label each expression on your number line.  
*Answers will vary. Refer to activity document.*



Students are provided with a kinesthetic entry point when they sort rational and irrational numbers and move to the designated side of the room.



Consider implementing the activity in the following way:

1. Distribute one card to each student.
2. Give students some time to independently complete question 1.
3. Identify two sides of the classroom—one rational and one irrational—and ask students to go to the side that matches their value. Students can compare with a partner on their side of the room to determine if they both agree. Allow students to correct their classification, if desired.
4. Have students split into groups based on the ranges indicated in the activity document.
5. Students work together in their group to compare and order values before plotting them on the number line.
6. Once they've reached consensus, they can copy their number lines into their workbooks, if desired.



Consider asking the following questions during a quick debrief after the activity.

- Did you or your peers make any mistakes during this activity? How did you correct them?
- What did you learn from these mistakes?



## 2.5.5 Bring It Together: Ordering Rational and Irrational Numbers

**Component Overview:** Students list helpful strategies for comparing, ordering, and plotting rational and irrational numbers on a number line.



### 2.5.5 Bring It Together: Ordering Rational and Irrational Numbers

1. Use your experience doing the previous activity to complete the following.
  - a. List any strategies you learned that are helpful to keep in mind when comparing and ordering rational and irrational numbers or plotting them on a number line.

| Strategies for Comparing and Ordering                                                                                                                                                                                                                                           | Strategies for Plotting on a Number Line                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><i>Answers will vary.</i><br/> <i>To plot the numbers on the number line, it's helpful to first find the value or approximate value of each. Putting all of the numbers into decimal form helps to be able to order them because they are all then in the same form.</i></p> | <p><i>Answers will vary.</i><br/> <i>To plot on the number line, it helps to make sure your numbers are in order whether least to greatest or greatest to least first. To plot the numbers on the number line, it helps to convert them all to decimals. It's also helpful to consider which two integers each value lies between and which of the two it is closer to.</i></p> |

- b. Compare and discuss your list with your partner. Add any strategies your partner shares that you find helpful that aren't on your list.

*Answers will vary.*



Encourage students to create their own sets of cards with a mixture of rational and irrational numbers between  $-25$  and  $25$ . Have students exchange sets of cards and order them from least to greatest.



## 2.5.6 Wrap-Up: Cool Down

**Component Overview:** Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



### 2.5.6 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.

*Answers will vary.*